Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

Exploratory Data Analysis using Excell

We began with data preparation and cleaning in Excel:

- **Data Loading:** Imported the dataset CSV file into Excel.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median of Review Rating.
- **Feature Engineering:**
  Created **age_group** column by binning customer ages.
  Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using MySQL Workbench (Business Transactions)

We performed structured analysis in MySQL Workbench to answer key business questions:

Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender	Total_Revanue
▶	Female	75191
	Male	157890

High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Customer_ID	Discount_Applied	Purchase_Amount
1	Yes	53
2	Yes	64
3	Yes	73
4	Yes	90
5	Yes	49
6	Yes	20
7	Yes	85
8	Yes	34
9	Yes	97
10	Yes	31
11	Yes	34
12	Yes	68
13	Yes	72
14	Yes	51
15	Yes	53
16	Yes	81
17	Yes	36

Top 5 Products by Rating – Found products with the highest average review ratings.

	Item_Purchased	average product rating
▶	Blouse	3.68
	Sweater	3.76
	Jeans	3.65
	Sandals	3.84
	Sneakers	3.76

Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	Shipping_Type	average_purchase_amount
▶	Express	60.48
	Standard	58.46

Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	Subscription_Status	total_customer	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
▶	Loyal	3116
	Returning	701
	New	83

Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group	total_revenue
▶	Middle-aged	89445
	Young-Adult	57279
	Adult	43193
	Senior	43164

Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



Business Recommendations

Boost Subscriptions – Promote exclusive benefits for subscribers.

Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

Review Discount Policy – Balance sales boosts with margin control.

Product Positioning – Highlight top-rated and best-selling products in campaigns.

Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.